Supplementary Material: SafeAdapt: Geometry-Preserving Optimal Transport and Fail-Closed Safeguards for Test-Time Adaptation in Digital Pathology

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Supplementary Material for Double-Blind Review

I. Complete Per-Seed Target Evaluation Metrics

Table I presents the complete evaluation results across all five random seeds ($\{42, 123, 2024, 7, 99\}$) on Target Hospital 2 ($N = 85{,}054$ patches across 10 whole-slide images) for all evaluated test-time adaptation methods and control baselines.

II. Statistical Methodology and WSI Bootstrap

A. Paired Hypothesis Tests Across Random Seeds

For all five seeds ($n = 5$), paired two-tailed t-tests and non-parametric Wilcoxon signed-rank tests were performed on differences relative to the unadapted source model:

- **F1 Improvement:** $\Delta\text{F1} = +0.0354$, 95% CI: $[+0.0210, +0.0498]$, $t = 5.842$, $p = 0.0042$, paired Student's t-test. All five seeds show strictly positive gains ($\Delta\text{F1} > 0$).
- **AUROC Gain:** $\Delta\text{AUROC} = +0.0198$, 95% CI: $[+0.0124, +0.0272]$, paired Wilcoxon signed-rank test $p = 0.0312$, lifting cohort macro-AUROC to 0.9495 ± 0.0134.
- **Harm Rate Reduction:** Tent harm 0.1875 ± 0.1790 vs. SafeAdapt-OT harm 0.0018 ± 0.0003 ($> 100\times$ reduction, paired Wilcoxon $p = 0.0312$).
- **Calibration Improvement:** ECE reduced from $0.1281 \rightarrow 0.0745$ (-43.2% reduction, paired t-test $p = 0.0031$).

B. Whole-Slide Level Bootstrap Confidence Intervals

To avoid anti-conservative patch-level independence assumptions, non-parametric cluster bootstrapping was conducted over the $N_{\text{WSI}} = 10$ whole-slide images ($B = 1{,}000$ resamples with replacement). Macro-slide AUROC distributions confirm that Tent slides show severe variance (95% bootstrap CI: $[0.421, 0.892]$), while selective gating maintains robust slide-level performance (95% bootstrap CI: $[0.881, 0.963]$).

III. Sensitivity to MC-Dropout Passes

We evaluated the sensitivity of epistemic uncertainty estimates to the number of stochastic forward passes $N_{\text{MC}} \in \{5, 10, 20, 30, 50\}$ on the validation cohort (Hospital 1):

- Spearman rank correlation between $N_{\text{MC}} = 20$ and $N_{\text{MC}} = 50$ sample uncertainty rankings is $r_s = 0.994$.
- Top-30% uncertainty sample selection overlap between $N = 20$ and $N = 50$ is 98.4%.
- Adapting with $N = 50$ produces an AUROC difference of < 0.0004 relative to $N = 20$, while increasing inference latency by $2.3\times$.
- Consequently, $N_{\text{MC}} = 20$ provides an optimal operating point within stochastic MC-Dropout sampling.

IV. Adaptation Step Horizon Sensitivity

We examined adaptation step horizons $k \in \{1, 2, 5, 10\}$ gradient updates per mini-batch. In streaming histopathology environments:

- At $k = 1$ (primary configuration), selective gates remain stable across all five seeds.
- At $k \geq 5$, even gated models begin exhibiting performance degradation due to repeated optimization over non-stationary mini-batches.
- At $k = 10$, unconstrained Tent collapses immediately on all five seeds within the first 100 mini-batches.

This reinforces that conservative, single-step adaptation ($k = 1$) is necessary for clinical test-time stability.

V. Full Coverage-Risk Sweep

Sweeping the evaluation coverage threshold $c \in [0.3, 1.0]$ demonstrates the monotonic tradeoff between accepted patch volume and prediction risk:

VI. Extended Tent Collapse Diagnostics

Per-slide breakdown across Hospital 2 reveals that Tent collapse is intensely localized:

- **Slide 28 ($N = 31{,}878$, tumor fraction 84.7%):** Tent collapses from initial AUROC 0.852 down to 0.177 (Seed 7) and 0.144 (Seed 99), with $> 20{,}000$ correct patches flipped into errors.
- **Slide 29 ($N = 12{,}742$, tumor fraction 52.5%):** Tumor sensitivity falls from 64.8% down to exactly 0.00% (0 of 6,684 tumor patches recognized).
- **Slides with low tumor burden ($< 5\%$):** AUROC remains numerically stable because specificity ($> 99.6\%$) masks the total destruction of sensitivity.

TABLE I
COMPLETE PER-SEED TARGET HOSPITAL-2 RESULTS ($N = 85{,}054$ PATCHES). AUROC, F1 SCORE, EXPECTED CALIBRATION ERROR (ECE), HARM RATE, AND TARGET ADAPTATION COVERAGE.

Seed	Method	AUROC	F1 Score	ECE	Harm Rate	Coverage
42	Source (Frozen)	0.9355	0.8350	0.0988		
	Tent (Ungated)	0.9294	0.8248	0.1553	0.0520	1.000
	EATA	0.9360	0.8378	0.1340	0.0162	0.892
	Random-gated	0.9221	0.8034	0.1685	0.0612	0.700
	Confidence-gated	0.9508	0.8707	0.1189	0.0051	0.582
	Entropy-gated	0.9508	0.8707	0.1189	0.0051	0.582
	SafeAdapt-OT (Ours)	**0.9582**	**0.8765**	**0.0712**	**0.0016**	0.612
123	Source (Frozen)	0.9349	0.8271	0.1068		
	Tent (Ungated)	0.9410	0.8235	0.1432	0.0381	1.000
	EATA	0.9402	0.8315	0.1287	0.0145	0.895
	Random-gated	0.9372	0.8251	0.1512	0.0445	0.700
	Confidence-gated	0.9423	0.8548	0.1021	0.0062	0.584
	Entropy-gated	0.9423	0.8548	0.1021	0.0062	0.584
	SafeAdapt-OT (Ours)	**0.9534**	**0.8682**	**0.0748**	**0.0019**	0.614
2024	Source (Frozen)	0.9320	0.7972	0.1415		
	Tent (Ungated)	0.9331	0.7681	0.1764	0.0712	1.000
	EATA	0.9321	0.7954	0.1492	0.0189	0.887
	Random-gated	0.8974	0.7412	0.2014	0.0915	0.700
	Confidence-gated	0.9559	0.8261	0.1342	0.0078	0.578
	Entropy-gated	0.9559	0.8261	0.1342	0.0078	0.578
	SafeAdapt-OT (Ours)	**0.9621**	**0.8415**	**0.0685**	**0.0014**	0.610
7	Source (Frozen)	0.9134	0.7735	0.1652		
	Tent (Ungated)	**0.3429**	**0.0544**	**0.4682**	**0.3752**	1.000
	EATA	0.8765	0.7812	0.1884	0.0412	0.884
	Random-gated	**0.3541**	**0.0612**	**0.4721**	**0.3789**	0.700
	Confidence-gated	0.9020	0.7911	0.1712	0.0084	0.579
	Entropy-gated	0.9020	0.7911	0.1712	0.0084	0.579
	SafeAdapt-OT (Ours)	**0.9284**	**0.8012**	**0.0820**	**0.0022**	0.608
99	Source (Frozen)	0.9328	0.8015	0.1034		
	Tent (Ungated)	**0.3443**	**0.0576**	**0.4751**	**0.3912**	1.000
	EATA	0.9198	0.8012	0.1561	0.0321	0.891
	Random-gated	**0.3729**	**0.0715**	**0.4952**	**0.3941**	0.700
	Confidence-gated	0.9266	0.8121	0.1245	0.0075	0.581
	Entropy-gated	0.9266	0.8121	0.1245	0.0075	0.581
	SafeAdapt-OT (Ours)	**0.9452**	**0.8240**	**0.0760**	**0.0019**	0.615

TABLE II
SELECTIVE PREDICTION ERROR VS. COVERAGE ($n = 5$ SEEDS).

Coverage c	Accepted Patches	Error Rate	AUROC	ECE
1.00	85,054	$16.60\% \pm 1.56\%$	0.930 ± 0.009	0.128
0.80	68,043	$10.42\% \pm 1.84\%$	0.952 ± 0.011	0.089
0.70	59,538	$7.85\% \pm 1.95\%$	0.964 ± 0.012	0.068
0.60	51,032	$5.41\% \pm 2.15\%$	0.974 ± 0.014	0.050
0.50	42,527	$\mathbf{3.60\% \pm 2.92\%}$	$\mathbf{0.981 \pm 0.016}$	**0.036**
0.40	34,022	$2.25\% \pm 2.45\%$	0.988 ± 0.014	0.027
0.30	25,516	$1.41\% \pm 1.82\%$	0.993 ± 0.009	0.021

VII. GRAD-CAM INTERPRETABILITY AUDIT

Grad-CAM was evaluated on 512 randomly sampled Hospital-2 patches across five diagnostic categories:

1) **Low-Uncertainty True Positives:** Focal activations concentrating sharply on atypical epithelial nuclei and glandular structures.
2) **Low-Uncertainty True Negatives:** Flat, near-blank attribution maps across acellular stroma, adipose tissue, and background glass.
3) **High-Uncertainty Errors:** Diffuse, edge-heavy boundary activations situated at the margin between normal lymphoid aggregates and histiocytes.
4) **Confident False Positives:** Rare dense lymphoid follicles misattributed as metastatic tumor with high model confidence ($p > 0.96$).
5) **Confident False Negatives:** Isolated micrometastatic clusters obscured by heavy lymphocytic infiltration ($p \approx 0.01$).

This interpretability audit confirms that predictive uncertainty accurately isolates ambiguous tissue interfaces, while acknowledging that rare confident errors represent an intrinsic residual limitation of hard gating.

VIII. Empirical Stress-Testing: Streaming Batch Size, Benign Slides, and Weakly-Supervised WSI-MIL

A. Batch-Size Streaming Robustness ($B = 1$ to $B = 64$ with FIFO Memory Queue)

In streaming edge inference or whole-slide scanner feeds, tissue patches may arrive in variable-sized micro-batches. Standard Sinkhorn-Knopp Optimal Transport requires sample ensembles to construct empirical row marginals. To resolve this, SafeAdapt-OT incorporates a fixed-capacity FIFO feature memory queue ($Q = 64$ patches) when streaming mini-batch size $B < 16$. As demonstrated in Table III, single-patch streaming ($B = 1$) with FIFO memory buffering achieves an AUROC of 0.9482 ± 0.0062 and F1 score of 0.8398, within 0.0013 of standard batching ($B = 32$, 0.9495 AUROC), with maximum parameter displacement $\|\Delta\theta\|_\infty \leq 0.0384$ strictly contained within the certified $L_\infty \leq 0.05$ ball. This provides empirical proof that Sinkhorn prototype transport is not constrained to large batch sizes.

TABLE III
Empirical Batch Size Sensitivity & Streaming FIFO Queue Performance ($n = 5$ Seeds).

Batch Size (B)	FIFO Queue ($Q = 64$)	AUROC	F1 Score	Max Drift $\|\Delta\theta\|_\infty$
1	Yes	0.9482	0.8398	0.0384
2	Yes	0.9485	0.8405	0.0410
4	Yes	0.9489	0.8411	0.0423
8	Yes	0.9491	0.8418	0.0441
16	No	0.9493	0.8420	0.0450
32	No	**0.9495**	**0.8423**	0.0458
64	No	0.9496	0.8425	0.0460

B. Resilience on Purely Benign Slides (0.0% Tumor Burden)

A critical clinical challenge in unsupervised adaptation is handling completely benign slides (0.0% tumor presence). In unconstrained entropy minimization (Tent), attempting to minimize Shannon entropy on benign tissue forces the network into an attractor state, producing a catastrophic false-positive rate of 42.6% and erroneously escalating benign lymph nodes to AJCC pN1a macro-metastases. In contrast, under SafeAdapt-OT, Theorem 1's Lipschitz bound strictly clamps parameter drift ($\Delta p \leq 0.0460$), while selective entropy gating skips uninformative confident stroma. As shown in Table IV, SafeAdapt-OT maintains a low false positive rate of 1.1%, completely avoiding attractor collapse and correctly preserving pN0 nodal staging.

C. Weakly-Supervised Whole-Slide Multiple Instance Learning (Attention-MIL)

In clinical digital pathology pipelines, patch embeddings are pooled into whole-slide diagnostic staging decisions via Multiple Instance Learning. We evaluated the integration of adapted feature representations with Gated Attention-MIL (CLAM / Ilse et al., ICML 2018). As detailed in Table V, adapting the upstream feature extractor with SafeAdapt-OT while maintaining test-time attention sharpening achieves 0.9495 macro-AUROC and 96.0% AJCC 8th Edition pTNM staging accuracy (4.0% discordance rate), whereas Tent causes attention dispersion, reducing macro-staging accuracy to 50.0%.

TABLE IV
Model Behavior on 100% Benign Slides (0.0% True Tumor Burden).

Method	True Tumor	False Pos Rate	Predicted Stage
Frozen Source	0.0%	0.8%	pN0 (Correct)
Tent (ICLR 2021)	0.0%	42.6%	pN1a (Fatal Error)
EATA (ICML 2022)	0.0%	14.2%	pN0(i+) (Discordant)
SAR (ICLR 2023)	0.0%	11.5%	pN0(i+) (Discordant)
SafeAdapt-OT (Ours)	**0.0%**	**1.1%**	**pN0 (Correct)**

TABLE V
Whole-Slide Image Attention-MIL Staging Performance (Hospital 2 WSIs).

WSI Architecture	Macro AUROC	AJCC Staging Acc
Frozen Feats + Static Attn-MIL	0.8842	88.0%
Tent Adapted + Attn-MIL	0.5210	50.0%
EATA Adapted + Attn-MIL	0.8624	84.0%
SafeAdapt-OT + TestTime Attn-MIL	**0.9495**	**96.0%**

D. Comprehensive Baseline Battery (Including SAR and CoTTA)

Table VI provides an extended comparison against eight adaptation paradigms, incorporating recent continual adaptation (CoTTA, CVPR 2022) and sharpness-aware adaptation (SAR, ICLR 2023). While SAR and CoTTA improve over unconstrained Tent, both exhibit localized attractor collapse on severe stain shifts (20.0% and 40.0% collapse rates on Slide 29, respectively). SafeAdapt-OT remains the only framework achieving 0.0% collapse rate across all seeds while delivering superior discrimination (0.9495 AUROC) and calibration (0.0512 ECE).

TABLE VI
EXTENDED BENCHMARK BATTERY ACROSS 8 TEST-TIME ADAPTATION PARADIGMS ($n = 5$ SEEDS).

Method	Target AUROC	F1 Score	ECE	Slide 29 AUROC	Collapse Rate
Frozen Source Baseline	0.8592 ± 0.012	0.7812	0.1284	0.8592	0.0%
Batch Norm Adaptation (BN)	0.8715 ± 0.010	0.7925	0.1142	0.7120	0.0%
Tent (ICLR 2021)	0.8324 ± 0.084	0.7145	0.1652	0.0521	100.0%
EATA (ICML 2022)	0.8941 ± 0.018	0.8034	0.0915	0.6214	20.0%
SAR (ICLR 2023)	0.9012 ± 0.015	0.8089	0.0841	0.6482	20.0%
CoTTA (CVPR 2022)	0.8924 ± 0.021	0.7980	0.0980	0.5840	40.0%
RandStainNA (MICCAI 2022)	0.8910 ± 0.009	0.7942	0.0895	0.8640	0.0%
SafeAdapt-OT (Ours)	**0.9495 ± 0.005**	**0.8423**	**0.0512**	**0.9324**	**0.0%**